Rubrics for Blueprint

The main objective is to enhance your proposal into a solid blueprint that you can refer back to throughout the entire course of your project.

Email your instructor (hbm3@queensu.ca or m.alam@queensu.ca) if you have any questions, and include your draft report in the email. **In your email, please list specific sections or concepts that you would like us to check. See guidelines below.**

Note:

- "Part I pdf" = "Part I: Requirements Hierarchy and Relationships"
- "Part II pdf" = "Part II: Requirements Hierarchy and Relationships Example"

Instructors

BP follows the proposal template, but include 10 pages, and the following.

Executive Summary (/5)

- is clear and concise.
- Motivation / problem is briefly mentioned without citing sources.
- gives brief project description, sequence of goals (design process), approach, scope, total cost.
- Just state the cost (and if you want, contingency budget) in executive summary, no need to include details about HST or other details.

Introduction (/5)

- Motivation and problem is expanded in slightly more detail from executive summary and appropriate peer-reviewed sources are chosen and cited.
- The objectives of the project are presented in a way that adequately address the stated problem(s).
- mentions options that could be reasonable to include in scope, but isn't due to time / resource constraints or different goals etc.
- includes general diagram of the system. Could be a physical diagram, system context diagram, or a higher-level logical or functional diagram.
- describes an intended operational scenario, including the system boundary, external interactions, environment, operating assumptions, and constraints.

Functional Description (/5 for everything except requirements section)

- includes general description and requirements specification section. 'Interface and Rerformance Specifications' has been changed to "Requirements Specifications" and the content has been changed accordingly.
- Design choices are justified (with context and adequate arguments) and alternatives are rejected with proper consideration of system needs, constraints, or other requirements.

- Functional decomposition diagrams are linked to logical decomposition diagrams (see slide 14 of Part I).
- **Prototypes** are ID'd and explicitly identified and characterized. (IDs are used for V&V section and milestones.)

Requirements Specifications Section (/5)

- Considerable thought was given to the organization of this section, in terms of breaking down the requirements into sub-requirements, subsystems, and so on.
- Functional decomposition diagram shows the interfaces and can trace those interfaces to interface requirements using proper labels.
- Critical requirements have diagrams (as shown in slide 10 of Part II pdf) showing relationships to functions, logical elements, sub-requirements, context, and system. Context for these critical requirements is adequately explained.
- Broad requirements are broken down into verifiable requirements, where context, justification, and proper references are included (see slide 7 of Part II pdf).
- are correctly-written 'shall' or 'may' statements. 'May' statements refer to stretch goals that students are not held accountable for in the final report.
- "shall not" statements are used to address potential faults, and are linked to V&V tests.
- Include functional requirements and non-functional requirements. Non-functional requirements include interface specs, performance specs, and any other requirements as necessary. See slide 15/20 in Part I pdf and research the relevant non-functional requirements for your system.
- are decomposed from system-needs into general requirements and into sub-requirements.
- are clear and readily understandable.
- are presented in a consistent way with each other and other parts of the report.
- are explained with adequate context. See slide 14 in Part II pdf.
- are verifiable, ID'd, and are each traceable to V&V tests and stakeholder needs.
- Critical, major, minor, and stretch goal requirements are specified.

Risks and Safety Hazards

- Context and reason for risk are identified
- Risk is defined precisely (cite a safety standard, SOP, or other published resource)
- There is a quantitative understanding of how large the risk is.
- Most of the risks I saw in proposals were too vague (e.g. "there are ethical risks")

Diagrams (/5)

- For functional decomposition, functions have been decomposed to a sufficient level of granularity that each lowest-level function performs a single transformation of energy, information, or material. At this level, the flows between functions only carry inputs and outputs, with no transformation of energy / information / material along the interfaces so that each component displayed is doing a conversion. Higher-level functional diagrams in the body with more detailed decompositions in the appendix are acceptable. However, at least one fully decomposed example should appear in the main body.
- The logical diagrams state what action is being performed between each element.
- In the physical diagram for physical hardware systems, it can be helpful to display components as they would visually appear, especially when components have a mechanical aspect (like propellors on

a drone, a ball moving across a surface, etc.)
- Flow-down in the diagrams is appropriate and specific to the project. Diagrams are not used as a way of displaying a series of bullet points that have little project-specific relation to one another. Rather, they organize the system into hierarchies, processes, and/or functions that link the low-level requirements to stakeholder needs.
- Typeset vector graphics (no low resolution or hand-drawn figures.)

Approach (/5)

- Includes more than 8 milestones that are explicitly ID'd and linked to specific tests or prototypes. Addresses division of labor. Specify whether using incremental or iterative approach, and this should be consistent with milestones. If relevant, detailed calculations are included in appendix.
- Contingencies involve any technical risks, safety risks, or time-limitation risks that could jeopardize the completion of any requirements specified in the previous section. Any contingency p lans show how the system and / or the approach changes as a result, and which milestone may be affected.
- Each **prototype** is referenced by its ID which was established in the functional description section and states: Purpose, scope, development approach, success criteria (requirements), traceability to tests and milestones.

Verification and Validation (V&V) Testing (/5)

- Evaluation sufficient for each component, and linked to requirements and specifications or intermediate prototypes.
- Diagram(s) included for experimental testing setup.
- Measureable acceptance criteria specified.
- If necessary, further tests are included in the appendix, but at least 5 tests should be included in the body of the report.
- All prototypes have explicit tests associated with each one.

Resource Requirements (/5)

- Milestones and relevant hardware traced appropriately.
- Budget is accurate. All entries have citations to the website or source from where they will be acquired.
- Total cost included.

Conclusion (/5)

- does not simply restate the executive summary or introduction. Instead, it:
- briefly returns to the problem from the introduction.
- summarizes the proposed solution and scope.
- shows that the requirements, functional description, V&V, and milestones form a consistent and traceable chain.
- defines what 'success' looks like for the project in terms of a general outline of deliverables.
- briefly assess feasibility in terms of budget, schedule, team skills, and other resources.
- discusses limitations of the project, such as: how it may be susceptible to failure, weaknesses or risks taken in the approach, dependencies that could compromise performance or availability,

assumptions that may not hold up in specific real-world environments or when scaling the system, edge-cases where the system may fail, etc.
- addresses the limitations' impact on satisfying requirements and / or stakeholder needs, and / or the main problem addressed in the introduction.
- suggests ways in which these limitations may be mitigated or addressed in future work.
- includes a statement about the team's current progress and the immediate next steps.

Formatting (/5)

- Max 10 pages for the body of the report (Introduction to Conclusion.)
- Formatting is correct in doc. Title page includes all information, such as your own logo for your project (replace the one in the template with your own design), Title, Group #, Group members, Supervisor, industry collaborator. Exec Summary page, then TOC page, List of Figures, List of Tables, and List of Acronyms. These lists must be complete. References cited in IEEE format.
- Font size is 12 pt for the main text (headings can be larger.) Body of report has 1.5 spacing. Table captions and figure captions are 1.0 spaced.

Writing (/5)

- doesn't clutter the document with sentences like "Eighteen tests, each with specific success criteria, are planned to confirm that all requirements are met." This is an example of a sentence that is too general, it is obvious because it states the definition of V&V testing, it takes up space, and adds little value to help the reader understand the specific project and specialized approach.
- The writing has been proofread to check if similarly vague sentences are present, which have been replaced with senteces that emphasize more critical aspects that are specific to the project.
- doesn't include the number of X that is present in the report (like X number of requirements / tests / subsystems, etc.) unless that number is especially significant, or you are listing them in a subsequent sentence.
- doesn't restate definitions of concepts that can be found in a textbook: "Each prototype is tested before the next one depends on it" (just say iterative approach).
- doesn't use figures that would be found in textbooks for systems engineering, since they are not specific to the project.
- doesn't discuss concepts that are obvious to someone grading capstone (e.g. that you are full-time students, or that this is an engineering project).
- Sentences that are broad relate back to the important concepts that are particular to that section, or to the project; doesn't use sentences like "this is the engineering work"; rather, the sentence relates back to the specific section back to the problem in the introduction, or the main point of that section.
- doesn't place emphasis where it is misleading to place emphasis.
- High specificity in language should be organized into the appropriate section, and not appear in a sentence that is discussing concepts more broadly, unless that specific concept is particularly critical to the project.
- Concepts or terms that are specific to the project are well-defined in the introduction, rather than being presented ad-hoc later in the report.
- Use consistent voice throughout. (e.g. first person.)

Citations (/5)

- The BP cites papers and sections of textbooks that provide the ideas for figures.

- Extensive use of peer-reviewed material for justifying decisions, such as scope, design, approach, etc. This includes requirements that are present in the appendix.
- The BP cites the websites or suppliers for all budget entries that will be purchased.
- If figures from other published sources are required to enhance the reader's understanding, these figures are not copied nor are they screenshots; rather, they are reproduced by hand, and permission is obtained from the author and publisher of the source to use them in the report.
- The BP uses the most reliable sources possible. Different claims call for different types of sources. See below:

Typical sources for requirements, specifications, compliance, constraints:

- The following are the authoritative sources for component ratings, protocol behavior, safety limits, and regulatory requirements:
- Standards (IEEE, ISO, IEC, etc.), manufacturer datasheets, reference manuals, etc.
- Do not use a paper or textbook as a citation here, unless the original document is not available.
- Cite the specific revision or version of the standard or datasheet used.

Fundamentals, Theory, Background

- Use textbooks for basic principles, derivations, and well-accepted methods (e.g., circuit laws, signal processing fundamentals, etc.).
- Review articles can be good sources for establishing state-of-the-art and justifying scope.
- Choose the type of textbook based on depth:
    - *Undergraduate textbooks* are appropriate for core concepts that the instructor and supervisor would know.
    - *Graduate-level textbooks* are good for more advanced and specialized topics.
    - *Engineering handbooks* are useful for standard practices in specialized fields.
- Use the most recent edition from established publishers.
- Cite the specific chapter or section, not just the textbook as a whole.

Design Decisions and Approach

- Use peer-reviewed literature, ranked from most to least reliable: journal articles > peer-reviewed conference papers > PhD dissertations > master's theses > preprints (e.g. arXiv) > grant proposals > previous capstone reports.
- Choose between experimental and theoretical paper based on the claim being supported:
    - Experimental papers are best for:
        - justifying methods that work in practice.
        - non-functional requirements, esp. performance specs.
        - comparing the specs of alternative or competing approaches.
    - Ensure that you identify how the experimental conditions differed in the paper compared to your own approach.
    - Theoretical papers are best for:
        - justifying a model, algorithm, derivation, or design principle.
        - establishing limits on what is achievable.
    - Theoretical papers should not be the only support for claims about real-world performance.
- Wherever possible, support your key design decisions with as much background as possible: theory (why the approach should work), experimental results to show that similar approaches do work in

subsystems or overall, and using review papers to show that the general idea fits in with other approaches in the field.

Websites

- Acceptable only from authoritative organizations, such as: manufacturers, suppliers, government agencies, standards bodies, and universities.
- Do not use blogs, forums, wikis, or videos as justification, but you can use them to find better sources.

Appendices (/5)

- Appendices are not used as a dumping ground for any figures or tables that couldn't fit into the 10 pages; these figures and tables should be simplified in the body of the report. The appendix provides only specific technical details that would otherwise detract from the crucial elements of the proposal.
- The appendices are referenced where needed, and are organized into sections that address the main body of the report.
- The appendices should not be needed to understand any aspect of the report, as many supervisors will not read them. (Instructors will.)

Supervisor and TA Rubric

Group number, Title

Introduction, Motivation & Scope (/5)

Are the problem, objectives, and scope: clear, appropriate, and feasible?

- Objectives align with the project, and address the main problem or motivation.
- Scope is feasible.
- Discussion of what is outside of scope is provided with context and is appropriate.
- Scope aligns with what was discussed in meetings with supervisor.

Functional Description & System Design (/5 for everything except requirements subsection)

Does the proposed system design appear technically appropriate and feasible for the project?

- System functions and interfaces are suitable for the intended application.
- Performance targets are realistic.
- The proposed design approach is technically sound.
- Design decisions are adequately justified. Alternatives are considered and shown to be inapplicable to the project due to specific constraints imposed by other design decisions, requirements, or stakeholder needs.
- Explain the system integration strategy, including subsystem dependencies, integration order, compatibility checks, and integration milestones.

Requirements Sub-Section (/5)

- are correctly-written 'shall' or 'may' statements. 'may' statements refer to stretch goals that students are not held accountable for in the final report.
- do not contradict each other. In addition, implicit and explicit assumptions made for each requirement do not contradict each other.
- do not contradict any other parts of the report, such as budget and milestones.
- are formed with valid assumptions.
- are feasible to achieve.
- do not have any ambiguity: each has only one interpretation.
- contain no factual errors.
- are unique and not redundant to other requirements, and are necessary to meet required functionalities and qualities.
- are complete: usable to form a set of "design-to" requirements.
- critical requirements are contained in the body of the report, and only further details (such as requirements for small subsystems) are contained in the appendix.

Development Approach & Milestones (/5)

Is the proposed development plan practical and appropriate for achieving the project objectives?

- Milestones are realistic and linked to meaningful tests or prototypes.
- Prototypes are developed with a detailed plan:
- Prototypes logically progress from low to high fidelity. Proof-of-concept prototypes address the parts of the project with the most risk early on. Each critical subsystem has at least one prototype that tests its operation on its own before integration. Subsystem boundaries are sensible, with the interfaces between them (signals, power, data, protocols) clearly defined so each prototype can be tested independently. Integrated prototypes are scheduled once their subsystems have passed standalone testing, and with enough buffer time for managing high risk development processes.
- Prototypes are described in sufficient detail, ID'd, and traced to V&V tests.
- The development approach is suitable for the project.
- No dependencies are missing.
- Contingencies and major risks are reasonably addressed.
- No large untested jumps between project stages.

Testing, Evaluation & Acceptance Criteria (/5)

Is the proposed testing and evaluation plan sufficient to demonstrate that the project meets its objectives?

- Tests are appropriate for the key system components.
- Testing is linked to requirements and specifications.
- Acceptance criteria are measurable and realistic.
- The proposed evaluation approach can demonstrate project success.
- Distinguish verification against technical requirements from validation against stakeholder needs and intended operation.

Resource Requirements & Budget (/5)

Are the proposed resources and budget appropriate and realistic for the project?

- Required hardware and resources are suitable for the planned work.
- Resources align with the project milestones.
- Estimated costs are reasonable.
- Total budget is realistic for the proposed scope.